

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	09 September 2026
Team ID	SWUID2026190387
Project Name	Medical Diagnosis Center by Service Portal (ServiceNow)
Maximum Marks	4 Marks

Technical Architecture:

The Medical Diagnosis Center by Service Portal is developed on the ServiceNow Platform using a 3-tier architecture. Patients interact with the Service Portal to register, book diagnostic tests, make payments, and download reports. ServiceNow processes the requests through Business Rules, Flow Designer, Notifications, and Script Includes. All patient, appointment, report, and payment data is stored securely in ServiceNow database tables

Architecture Diagram

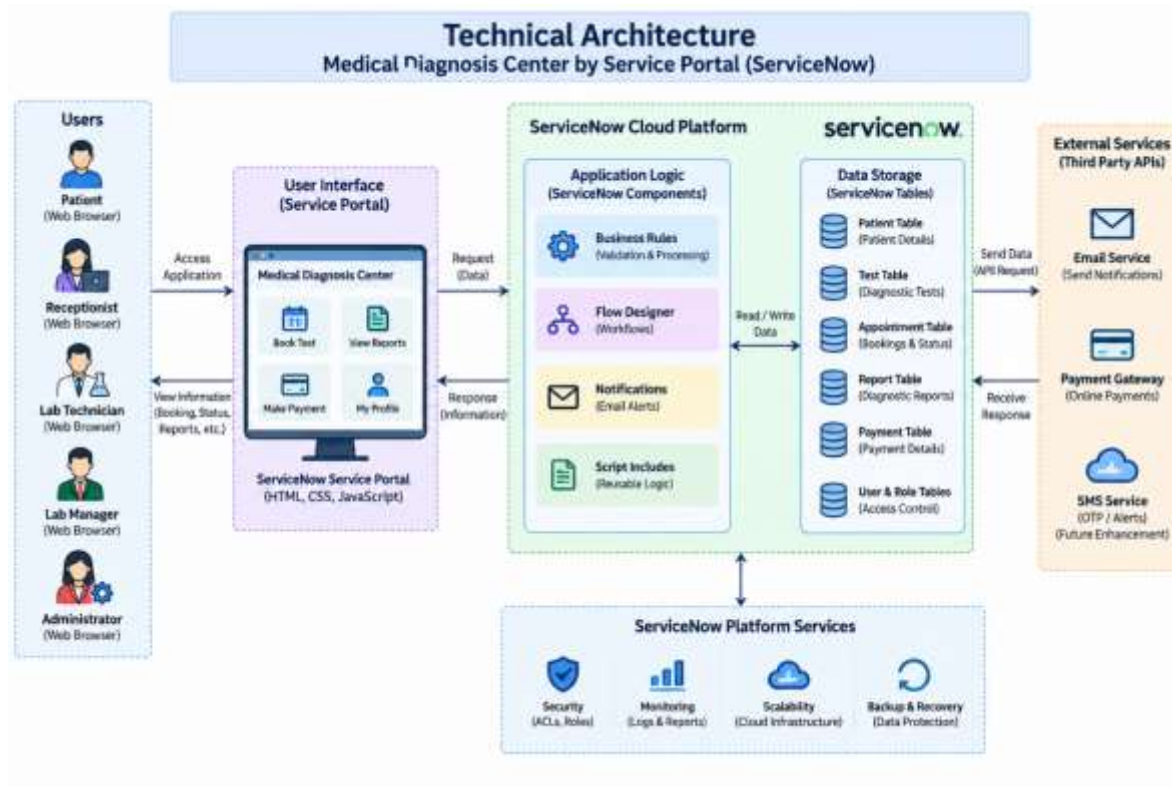


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Patients and staff interact with the application through the Medical Diagnosis Center Service Portal.	ServiceNow Service Portal, HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic- Appointment Management	Handles patient registration, appointment booking, slot validation, and booking rules.	Business Rules, Client Scripts, Script Includes
3.	Application Logic- Workflow Automation	Manages appointment lifecycle, approval workflow, report processing, and notifications.	Flow Designer, Notifications, Email Templates
4.	Application Logic- Payment Management	Calculates fees, tracks payment status, generates invoices, and manages refunds.	Business Rules, Flow Designer, Currency Fields
5.	Database	Stores patient details, diagnostic tests, appointments, reports, and payment information.	ServiceNow Custom Tables (Scoped Application)
6.	Cloud Database	Centralized storage for all records within the ServiceNow platform.	ServiceNow Cloud Database
7.	File Storage	Stores uploaded diagnostic reports and related documents securely.	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API- Email Service	Sends booking confirmation, payment confirmation, approval, and report availability emails.	ServiceNow Attachments
9.	External API- Payment Integration	Supports secure online payment processing and invoice generation.	Payment Gateway Integration (Future Phase)
10.	Machine Learning Model	Not used in the current project implementation.	Not Applicable (N/A)
11.	Infrastructure (Server / Cloud)	Deploys the application on the ServiceNow cloud platform.	ServiceNow Cloud Platform (Scoped Application)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frontend widgets and portal pages are customized using ServiceNow web technologies.	HTML, CSS, JavaScript, Service Portal Widgets
2.	Security Implementations	Protects patient records using login authentication, ACLs, role-based access control, and secure permissions.	ACLs, Roles, ServiceNow Authentication, RBAC
3.	Scalable Architecture	The application supports adding more patients, diagnostic tests, appointments, reports, and users without changing the overall architecture.	Scoped Application, ServiceNow Cloud Architecture
4.	Availability	The Service Portal is available online for patients and hospital staff anytime through the ServiceNow cloud platform.	ServiceNow Cloud Infrastructure
5.	Performance	Real-time slot validation, optimized business rules, and Flow Designer automation provide faster appointment processing and report updates.	Business Rules, Flow Designer, ServiceNow Database

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>